1. Problem 1

$$
\begin{aligned}
P &= \frac{2\pi^2 k_B^4}{15c^2h^3}T^4 \\
\text{SA} &= 0.65\ \text{m}^2 \\
T &= 34°\text{C} = 307.15\ \text{K} \\
P &= \epsilon\sigma T^4 \\
P &= 1 \cdot 5.670 \cdot 10^{-8}\ \text{J/m}^2\text{/s/K}^4 \cdot (307.15\ \text{K})^4 \\
P &= 504.6\ \text{J/m}^2\text{/s} \\
P &= 504.6\ \text{J/m}^2\text{/s} \cdot 0.65\ \text{m}^2 \\
&\boxed{P = 328.02\text{W}}
\end{aligned}
$$

2. Problem 2 (2.5)

$$
\begin{aligned}
\lambda_{\max} &= \frac{hc}{5k_BT} \\
h &= \frac{\lambda_{\max}5k_BT}{c} \\
\lambda_{\max} &= \frac{c_2}{5T} \\
c_2 &= 1.434 \cdot 10^{-2} \\
k_B &= 1.38 \cdot 10^{-23} \text{ J/K} \\
c &= 3 \cdot 10^{8} \text{ m/s}
\end{aligned}
$$

T (K)	$\lambda_{\max}$	$h = \frac{\lambda_{\max}5k_BT}{c}$
1646	$1.75 \cdot 10^{-6}$	$6.626 \cdot 10^{-34}$ J/s
1449	$2.1 \cdot 10^{-6}$	$6.998 \cdot 10^{-34}$ J/s
1259	$2.3 \cdot 10^{-6}$	$6.66 \cdot 10^{-34}$ J/s

$\boxed{h_{\text{avg}} = 6.76 \cdot 10^{-34} \text{ J/s}}$

3. Problem 3 (2.7)

$$
\begin{aligned}
< \nu > &= \frac{3N_a h\nu}{\exp\left(h\nu/\left(k_B T\right)\right) - 1} \\
\nu &= \frac{c}{\lambda} \\
h &= 6.626 \cdot 10^{-34} \\
c &= 3 \cdot 10^{8} \\
k_B &= 1.38 \cdot 10^{-23} \\
T &= 1635 \text{ K} \\
k_B \cdot T &= 2.258 \cdot 10^{-20}
\end{aligned}
$$

λ (μm)	ν s^{-1}	$h\nu$	$\frac{h\nu}{k_B T}$ (J)	$< \nu >= \frac{3N_a h\nu}{\exp(h\nu/(k_B T))-1}$ (J)
1	$2.998 \cdot 10^{14}$	$1.986 \cdot 10^{-19}$	8.797	$3.002 \cdot 10^{-23}$
2	$1.499 \cdot 10^{14}$	$9.932 \cdot 10^{-20}$	4.399	$1.236 \cdot 10^{-21}$
5	$5.996 \cdot 10^{13}$	$3.973 \cdot 10^{-20}$	1.759	$8.261 \cdot 10^{-21}$
20	$1.499 \cdot 10^{13}$	$9.932 \cdot 10^{-19}$	0.440	$1.798 \cdot 10^{-20}$

The wavelength of 1 μm has the smallest energy. Therefore, at this wavelength, most of the photons will have zero energy.

4. Problem 4 (2.9)

   Average Vibrational energy

   S1 T = 400K $\nu = 12 \cdot 10^{12}$ s^{-1}

   a)

$$
\begin{aligned}
< \nu_{\text{vib}} > &= \frac{3N_a h\nu}{\exp\left(h\nu / \left(k_B T\right)\right) - 1} \\
< \nu_{\text{vib}} > &= \frac{3 \cdot 6.02 \cdot 10^{23} \cdot 6.626 \cdot 10^{-34} \cdot 12 \cdot 10^{-12}}{\exp\left[6.626 \cdot 10^{-34} \cdot 12 \cdot 10^{-12} / \left(1.38 \cdot 10^{-23} \cdot 400\right)\right] - 1} \\
&\boxed{< \nu_{\text{vib}} >= 4457 \text{ J/mol}}
\end{aligned}
$$

   b)

   $\boxed{3N_a h\nu = 14364 \text{ J/mol}}$

5. Problem 5 (2.11)

$$\begin{aligned}
\lambda\nu &= c \\
\nu &= \frac{c}{\lambda} \\
c &= 3 \cdot 10^8 \text{ m/s} \\
1 \text{ eV} &= 1.602 \cdot 10^{-19} \text{ J}
\end{aligned}$$

Convert wavelength to frequency and kinetic energy from eV to J.

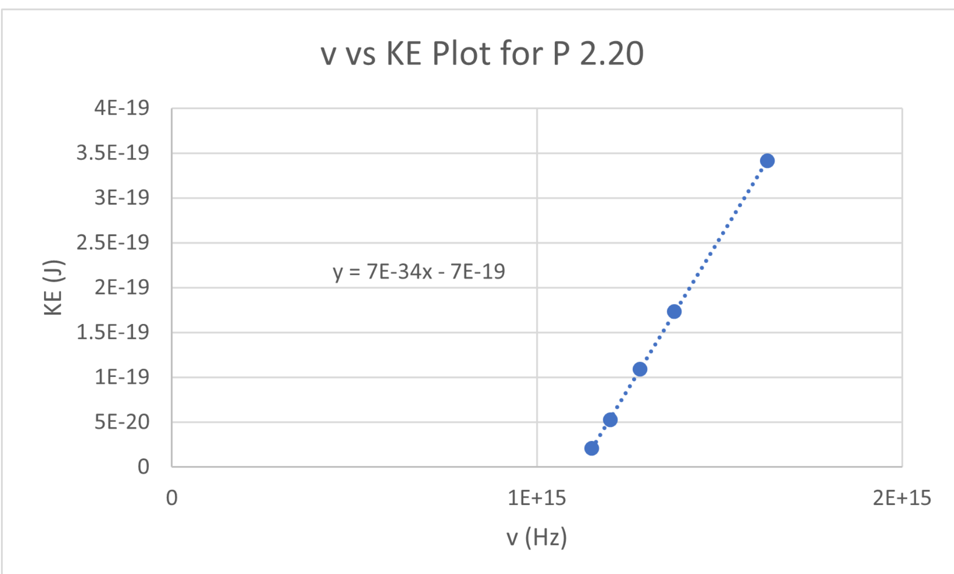
Tabulated values:

λ (nm)	KE (eV)	ν (Hz)	KE (J)
261	0.13	$1.15 \cdot 10^{15}$	$2.08 \cdot 10^{-20}$
250	0.33	$1.20 \cdot 10^{15}$	$5.29 \cdot 10^{-20}$
234	0.68	$1.28 \cdot 10^{15}$	$1.09 \cdot 10^{-19}$
218	1.08	$1.38 \cdot 10^{15}$	$1.73 \cdot 10^{-19}$
184	2.13	$1.63 \cdot 10^{15}$	$3.41 \cdot 10^{-19}$

$$E_k = h\nu - \phi$$

Fitting the data to the above equation yields:

$$E_k = 6.68 \cdot 10^{-34} \cdot \nu - 7.48 \cdot 10^{-19}$$



$\boxed{h = 6.68 \cdot 10^{-34} \text{ J·s}}$

$\phi = 7.48 \cdot 10^{-19}$

$\boxed{\phi = 4.67 \text{ eV}}$

6. Problem 6 (2.20)

a.

$$\begin{aligned}
\mathrm{MW}_{\mathrm{H_2}} &= 2.016\ \mathrm{g/mol} \\
\mathrm{m}_{\mathrm{H_2}} &= \frac{2.016\ \mathrm{g/mol}}{6.022 \cdot 10^{23}\ \mathrm{mol}^{-1}} \cdot \frac{1\ \mathrm{kg}}{1000\ \mathrm{g}} \\
\mathrm{m}_{\mathrm{H_2}} &= 3.35 \cdot 10^{-27}\ \mathrm{kg}
\end{aligned}$$

Photon momentum

$$p = \frac{h}{\lambda}$$

Particle momentum

$$p = mv$$

Momentum is conserved

$$\begin{aligned}
mv &= \frac{h}{\lambda} \\
v &= \frac{h}{\lambda m} \\
v &= \frac{6.62 \cdot 10^{-34}\ \mathrm{J{\cdot}s}}{5.84 \cdot 10^{-8}\mathrm{m} \cdot 3.35 \cdot 10^{-27}\ \mathrm{kg}} \\
&\boxed{v = 3.39\ \mathrm{m/s}}
\end{aligned}$$

b.

$$\begin{aligned}
\mathrm{KE} &= \frac{1}{2} m v^2 \\
\mathrm{KE} &= \frac{1}{2} \cdot 3.35 \cdot 10^{-27}\ \mathrm{kg} \cdot (3.39\ \mathrm{m/s})^2 \cdot 6.022 \cdot 10^{23}\ \mathrm{mol}^{-1} \\
&\boxed{\mathrm{KE} = 1.16 \cdot 10^{-2}\ \mathrm{J/mol}}
\end{aligned}$$